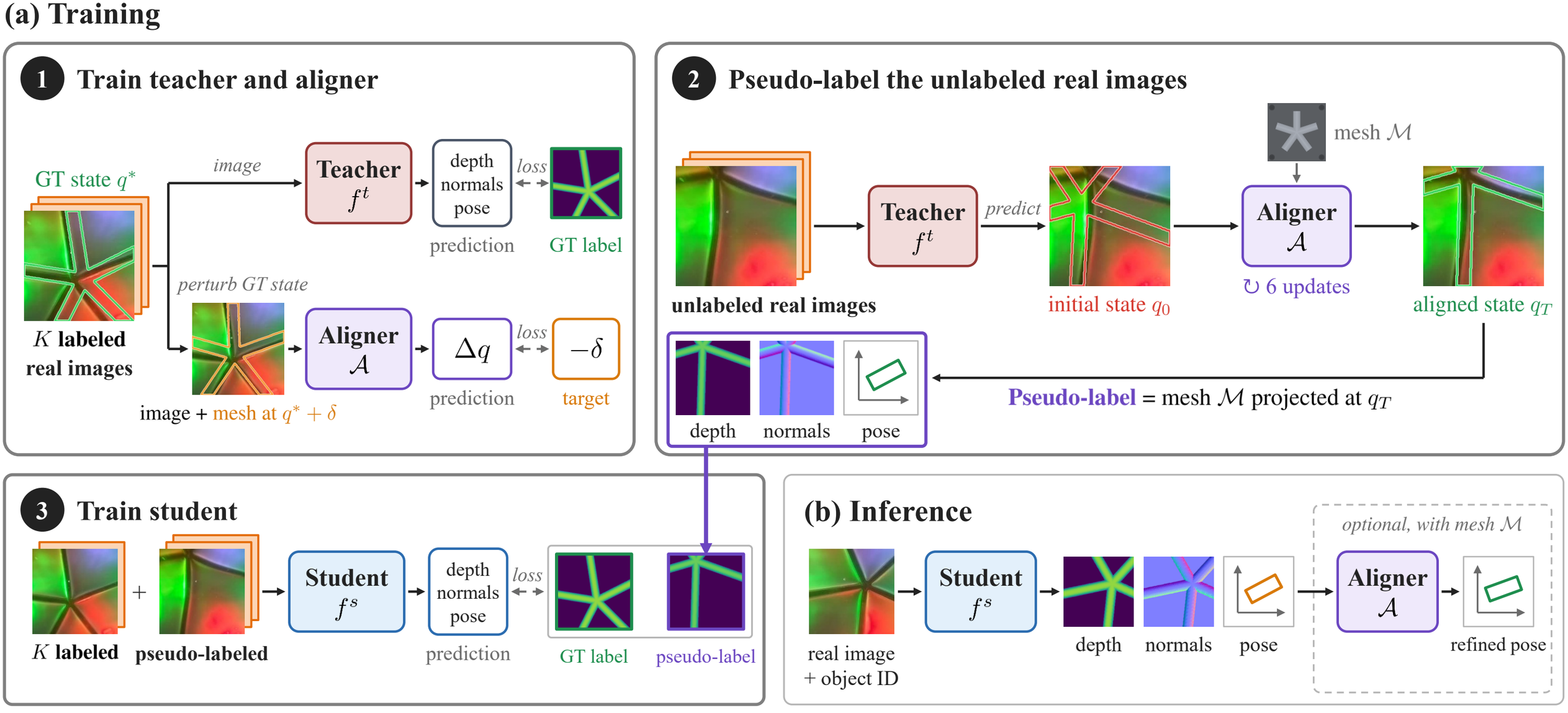

(a) Training
1 Train teacher and aligner
GT state q^*
image
Teacher f^t
depth
normals
pose
loss
prediction
GT label
K labeled real images
perturb GT state
Aligner $\mathcal{A}$
Δq
$-\delta$
target
image + mesh at $q^* + \delta$
2 Pseudo-label the unlabeled real images
mesh $\mathcal{M}$
unlabeled real images
Teacher f^t
predict
initial state q_0
Aligner $\mathcal{A}$
↻ 6 updates
aligned state q_T
depth
normals
pose
Pseudo-label = mesh $\mathcal{M}$ projected at q_T
3 Train student
K labeled
pseudo-labeled
Student f^s
depth
normals
pose
loss
prediction
GT label
pseudo-label
(b) Inference
optional, with mesh $\mathcal{M}$
real image
+ object ID
Student f^s
depth
normals
pose
Aligner $\mathcal{A}$
refined pose